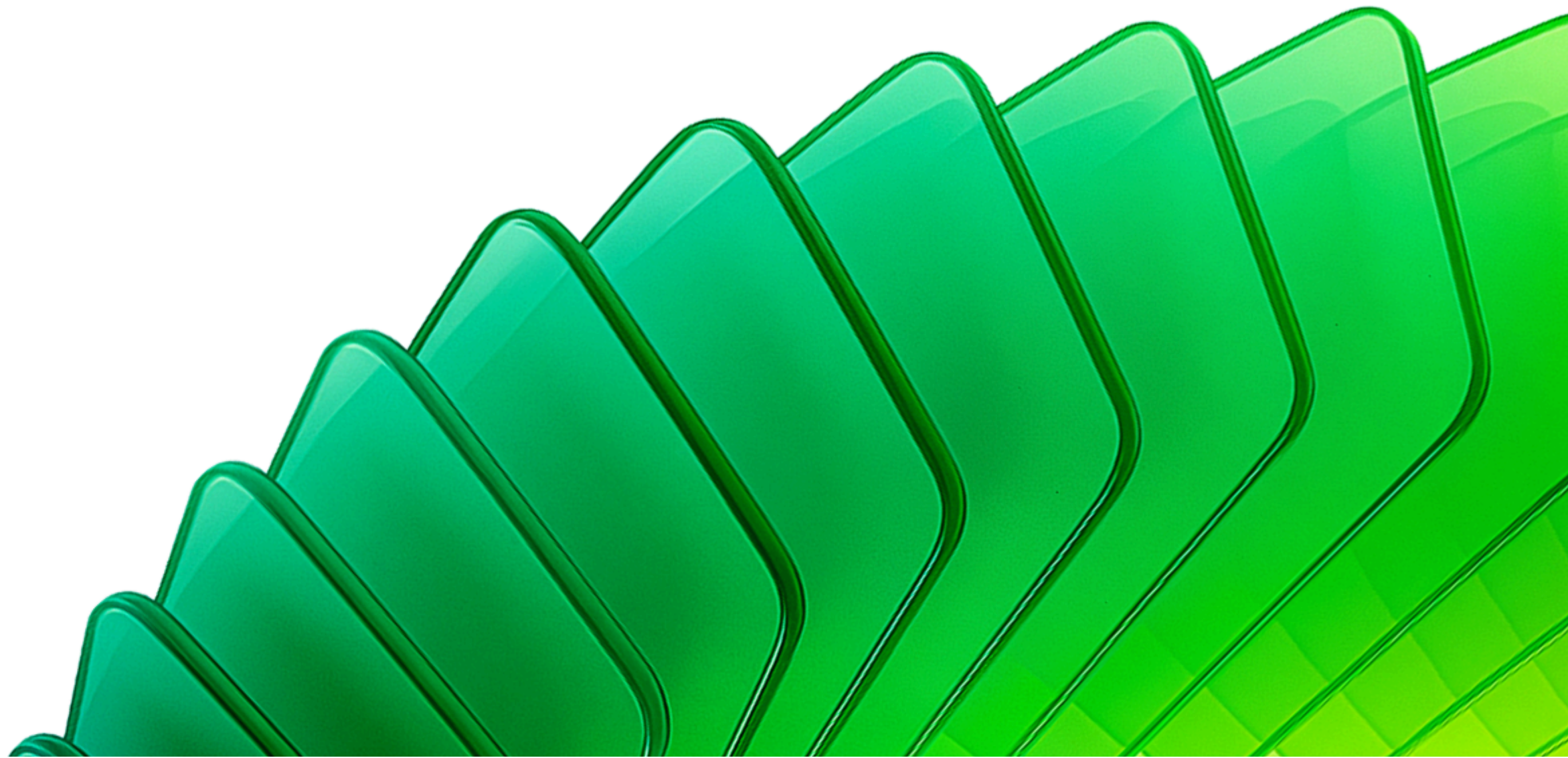


# Gen AI Academy

APAC Edition

Build in APAC. Build for the world.



## PARTICIPANT DETAILS

**Participant Name** - Hrituraj Saha.

**Problem Statement** - Build a multi-agent AI system that helps users manage tasks, schedules and information by interacting with multiple tools and data sources.

## BRIEF ABOUT THE IDEA

**Gymnasium AI Agent** is a multi-agent AI-powered fitness assistant designed to help users manage workouts, track progress, and receive personalized exercise guidance through natural language interaction. The system leverages a multi-agent architecture, where a central (root) agent coordinates with a specialized Gym Coach agent to understand user intent and execute tasks. These tasks are carried out using integrated tools via the Model Context Protocol (MCP), enabling real-time actions such as logging workouts, tracking fitness progress, and retrieving historical data.

Unlike traditional fitness apps, Gymnasium Agent goes beyond static tracking by enabling intelligent, multi-step workflows. For example, a user can log a workout, record their weight, and retrieve progress insights—all within a single interaction. The system is backed by persistent storage (Google Cloud Datastore), ensuring that all user data such as workouts and fitness logs are securely stored and retrievable across sessions. Deployed as a FastAPI-based service, the agent can be easily integrated into web or mobile applications, making it scalable and production-ready.

Overall, Gymnasium Agent demonstrates how AI agents can evolve from simple chat interfaces into action-driven systems capable of reasoning, planning, and executing real-world tasks in the fitness domain.

# SOLUTION APPROACH & PROBLEM TRANSLATION

The given problem required building a multi-agent AI system capable of managing tasks, interacting with tools, handling workflows, and storing structured data.

To address this, the solution was designed using a modular agent-based architecture -

- Google ADK (Agent Development Kit) was used to define and orchestrate multiple agents.
- A Root Agent handles user intent and delegates execution.
- A specialized Gym Coach Agent performs domain-specific actions (fitness tracking & guidance).
- Sequential Agent workflow ensures structured execution of multi-step tasks.

For tool integration, MCP (Model Context Protocol) was implemented -

- Enabled seamless connection between agents and tools.
- Tools such as workout logging and progress tracking were exposed as callable functions.
- Ensured real-time execution of user requests.

For persistence, Google Cloud Datastore was used -

- Stores workouts and fitness logs.
- Enables retrieval of historical data across sessions.
- Provides structured, scalable storage.

This architecture ensures all core hackathon requirements are met -

- Multi-agent coordination.
- Tool integration.
- Persistent memory.
- Multi-step workflow execution.
- API-based deployment.

# REAL-WORLD PROBLEM & IMPACT

## Problem

Most fitness applications today are -

- Static and non-interactive.
- Require manual tracking.
- Lack personalization and intelligent guidance.
- Do not adapt to user progress dynamically.

Users often struggle with -

- Maintaining consistency.
- Planning workouts.
- Tracking progress effectively.

## Solution & Impact

Gymnasium Agent solves this by acting as an AI-powered personal fitness assistant -

- Users can interact using natural language.
- Workouts and progress are automatically logged and tracked.
- The system provides context-aware suggestions and guidance.
- Enables seamless, hands-free fitness management.

## Practical Impact

- Improves user consistency and accountability.
- Reduces friction in tracking fitness data.
- Provides personalized fitness insights.
- Demonstrates real-world applicability of agentic AI systems.

# CORE WORKFLOW OF THE SOLUTION

The system follows a structured multi-agent execution pipeline -

## 1. User Input

- User provides a natural language prompt.
- Example - “Log pushups 15 reps 3 sets”.

## 2. Root Agent (Intent Handling)

- Captures user input.
- Stores it in shared state.
- Delegates execution to workflow agent.

## 3. Gym Coach Agent (Execution Layer)

- Interprets user intent.
- Decides which tools to use.
- May perform multi-step reasoning.

## 4. MCP Tools (Action Layer)

- Executes specific operations -
  - Add workout.
  - Retrieve history.
  - Log fitness progress.

## 5. Database (Persistence Layer)

- Stores structured data.
- Ensures memory across sessions.

## 6. Response Generation

- Agent returns a contextual, human-like response.

The system demonstrates how -

**AI agents can reason → Plan → Act → Store → Retrieve**

# OPPORTUNITIES & DIFFERENTIATION

## How is this different from existing solutions?

Most existing fitness applications and trackers are -

- Static and form-based (manual input required).
- Limited to tracking without intelligent reasoning.
- Do not support multi-step workflows.
- Lack real-time adaptability and contextual understanding.

In contrast, Gymnasium Agent introduces a fully agentic approach -

### 1. Natural Language Interface

- Users interact conversationally instead of filling forms.

### 2. Multi-Agent Intelligence

- Instead of a single AI model, the system uses -
  - A coordinating Root Agent.
  - A specialized Gym Coach Agent.

### 3. Tool-Driven Execution (MCP)

- The agent doesn't just respond—it takes actions -
  - Logs workouts.
  - Tracks progress.
  - Retrieves data.

### 4. Workflow Automation

- Handles multi-step tasks in a single request (e.g., log workout + track weight + show history).

### 5. Persistent Memory

- Unlike typical chatbots, it remembers user data across sessions.

# UNIQUE SELLING PROPOSITION (USP)

## 1. Action-Oriented AI

The system goes beyond conversation -

- It executes real tasks using tools and stores results.

## 2. True Multi-Agent System

- Clear separation of responsibilities.
- Scalable architecture.
- Demonstrates real-world agent orchestration.

## 3. MCP-Based Tool Integration

- Standardized way to connect AI with external tools.
- Makes the system extensible and production-ready.

## 4. Multi-Step Reasoning & Execution

- Understands complex user requests.
- Breaks them into actionable steps.
- Executes them sequentially.

## 5. Memory-Driven Personalization

- Tracks workouts and progress over time.
- Enables contextual and personalized responses.

## 6. API-First Scalable Design

- Easily integrable into mobile/web apps.
- Suitable for real-world deployment.

# LIST OF FEATURES OFFERED BY THE SOLUTION

## Core AI Features

- **Multi-Agent Architecture**
  - Root Agent for orchestration.
  - Gym Coach Agent for domain-specific execution.
- **Natural Language Interaction**
  - Users can log workouts and query progress conversationally.
- **Context-Aware Responses**
  - Understands user intent and provides relevant actions or suggestions.
  -

## Functional Features

### 1. Workout Management

- Log workouts (exercise, reps, sets).
- View complete workout history.
- Structured storage of workout sessions.

### 2. Fitness Progress Tracking

- Log body weight and fitness notes.
- Retrieve progress history.
- Enables tracking of improvements over time.

### 3. Multi-Step Task Execution

- Handles compound requests in one interaction.
  - Example - log workout + track weight + show history.

# LIST OF FEATURES OFFERED BY THE SOLUTION

## 4. Intelligent Recommendations

- Suggests workout plans.
- Provides exercise guidance.
- Can recommend routines based on user goals.

## Integration Features

- **MCP (Model Context Protocol) Integration**
  - Seamless tool invocation by agents.
  - Standardized interaction between AI and tools.
- **API-Based Access**
  - REST endpoint using FastAPI.
  - Easy integration with frontend or external apps.

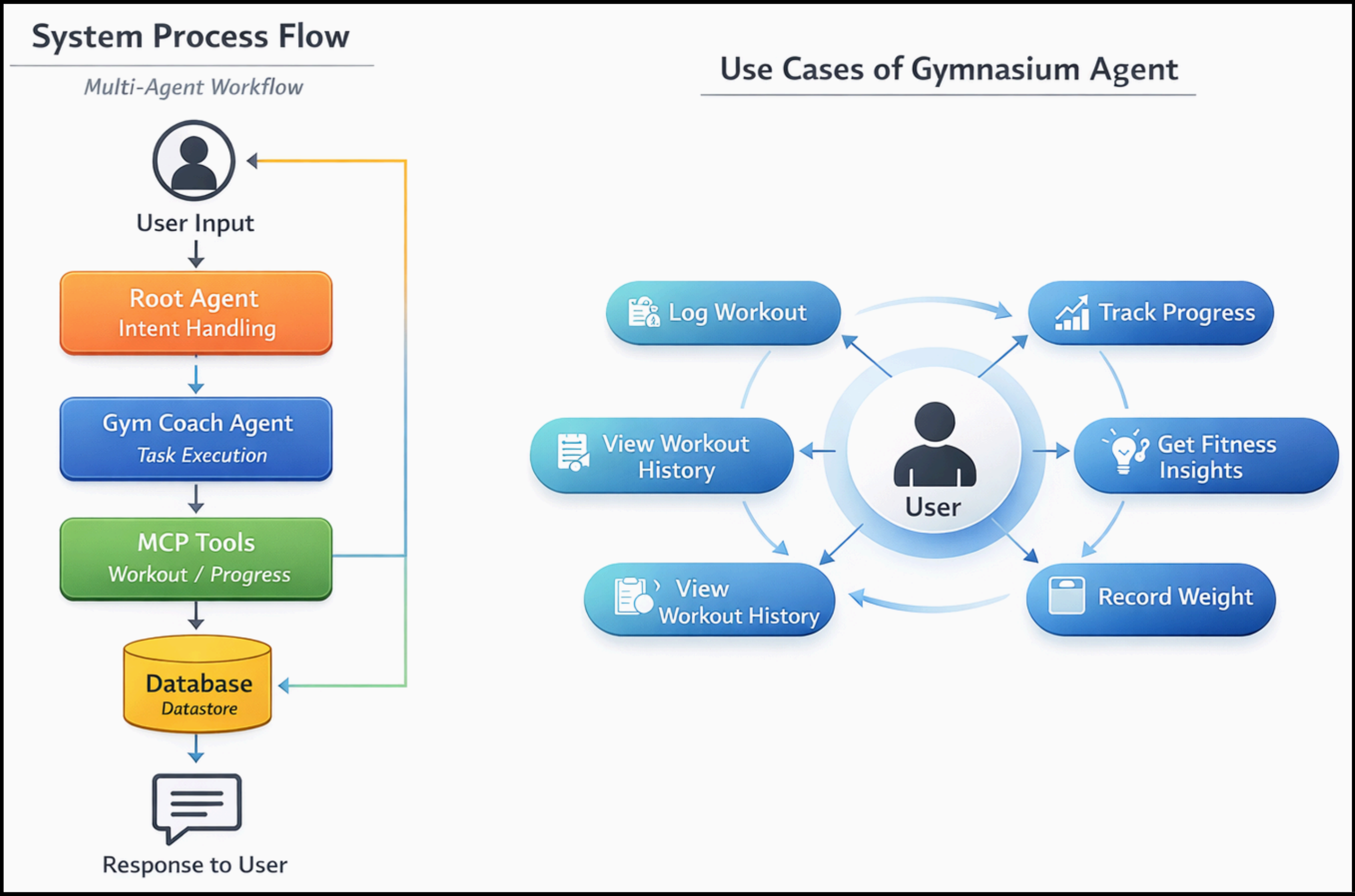
## Data & Memory Features

- **Persistent Storage**
  - Uses Google Cloud Datastore.
  - Stores workouts and progress logs.
- **Cross-Session Memory**
  - Retains user data across interactions.

## System Capabilities

- **Real-Time Execution**
  - Immediate tool invocation and response.
- **Scalable Architecture**
  - Modular agent design for future expansion.
- **Error Handling**
  - Graceful handling of invalid inputs.

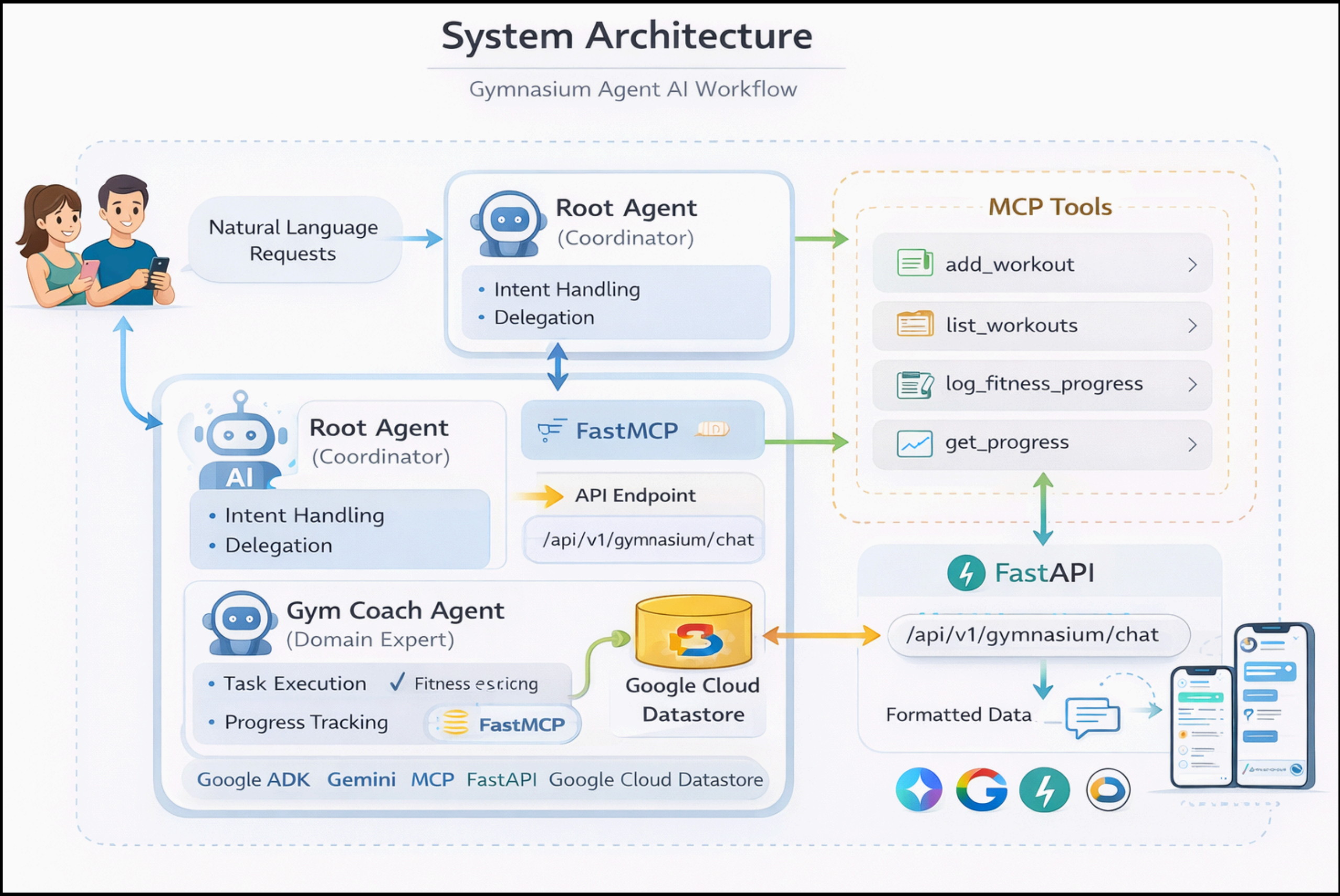
# PROCESS FLOW DIAGRAM OR USE-CASE DIAGRAM



# WIREFRAMES/MOCK DIAGRAMS OF THE PROPOSED SOLUTION



# ARCHITECTURE DIAGRAM OF THE PROPOSED SOLUTION



# TECHNOLOGIES/GOOGLE SERVICES USED IN THE SOLUTION

## Core Technologies

### 1. Google ADK (Agent Development Kit)

- Used to build and orchestrate multi-agent workflows.
- Enables modular, scalable agent design.

### 2. MCP (Model Context Protocol) – FastMCP

- Connects agents with tools (workout logging, progress tracking).
- Standardized and extensible tool integration.

### 3. FastAPI

- Backend framework for API deployment.
- High performance, async, production-ready.

### 4. Gemini (LLM)

- Powers reasoning, understanding, and responses.
- Efficient and strong for real-time AI tasks.

## Google Cloud Services

### 1. Google Cloud Datastore

- Stores workouts and fitness logs.
- Scalable, reliable, persistent memory.

### 2. Google Cloud Logging

- Tracks errors and system activity.
- Ensures monitoring and debugging.

# TECHNOLOGIES/GOOGLE SERVICES USED IN THE SOLUTION

## Why This Stack?

- Modular Architecture → Easy to extend and maintain.
- Tool-Based Execution (MCP) → Enables real-world actions.
- Persistent Storage → Supports personalization and memory.

## Scalability & Deployment

- API-first design → Easy integration with apps.
- FastAPI → Handles concurrent users.
- MCP → Add new tools/agents easily.
- Cloud DB → Scales with user growth.

## Key Takeaway

Combines AI reasoning + tool execution + scalable backend to build a real-world, deployable intelligent system.

# SNAPSHOTS OF THE PROTOTYPE

Agent Development Kit

gymnasium\_agent

TraceEventsStateArtifactsSessionsEval

Invocations

Hello

I did 10 pushups for 3 sets today

Add squat workout 20 reps 4 sets

Log today's workout and show my history

Suggest a beginner workout plan

What muscle groups am I missing?

SESSION ID dfce0134-1915-4379-a132-9e35537bba36

Token StreamingNew Session

Hello

⚡ transfer\_to\_agent

✓ transfer\_to\_agent

Hello there! I'm your AI fitness coach, here to help you crush your goals. How can I assist you today? We could start by logging a workout or checking your progress!

I did 10 pushups for 3 sets today

⚡ transfer\_to\_agent

✓ transfer\_to\_agent

⚡ add\_workout

✓ add\_workout

Fantastic work! Logging 3 sets of 10 pushups is a great start. Consistency is key! What's next for your fitness journey?

Add squat workout 20 reps 4 sets

⚡ transfer\_to\_agent

✓ transfer\_to\_agent

⚡ add\_workout

✓ add\_workout

Type a Message...

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Activate Windows  
Go to Settings to activate Windows.

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Token Streaming | + New Session

Great job adding squats to your workout! 4 sets of 20 reps is excellent for building strength. Keep up the amazing work! Is there anything else I can help you log or track today?

Log today's workout and show my history

⚡ transfer\_to\_agent

✓ transfer\_to\_agent

⚡ list\_workouts

✓ list\_workouts

Here's your workout history:

- **pushups:** 3 sets x 10 reps
- **squat:** 4 sets x 20 reps

You've been consistent and that's fantastic! Keep up the great work! If you have another workout to add or want to check something else, just let me know!

Suggest a beginner workout plan

⚡ transfer\_to\_agent

✓ transfer\_to\_agent

Absolutely! Here's a great beginner full-body workout plan you can do 2-3 times a week, with a day of rest in between:

**Warm-up (5-10 minutes):**

- Light cardio (jogging in place, jumping jacks)
- Dynamic stretches (arm circles, leg swings)

**Workout:**

1. **Squats:** 3 sets of 10-12 reps (bodyweight or light weights)
2. **Push-ups:** 3 sets of as many reps as you can do (on knees or toes)

Type a Message...

Activate Windows  
Go to Settings to activate Windows.

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Warm-up (5-10 minutes):

Light cardio (jogging in place, jumping jacks)

Dynamic stretches (arm circles, leg swings)

Workout:

1. Squats: 3 sets of 10-12 reps (bodyweight or light weights)

2. Push-ups: 3 sets of as many reps as you can do (on knees or toes)

3. Lunges: 3 sets of 10-12 reps per leg (bodyweight)

4. Dumbbell Rows (or Bent-Over Rows with resistance band): 3 sets of 10-12 reps

5. Plank: 3 sets, hold for 20-30 seconds

6. Glute Bridges: 3 sets of 12-15 reps

Cool-down (5-10 minutes):

Static stretches (hold each stretch for 20-30 seconds)

Remember to focus on proper form, and it's okay to start slow and gradually increase your reps, sets, or resistance as you get stronger. Let me know when you try it, and we can log your progress!

What muscle groups am I missing?

transfer\_to\_agent

transfer\_to\_agent

That's a great question for optimizing your routine!

Based on the beginner full-body plan I suggested, which is quite comprehensive for starting out, here are a few muscle groups that could receive more *direct* or *isolated* attention as you progress:

1. Lateral/Side Deltoids (Shoulders): While push-ups and rows engage your shoulders, dedicated exercises like lateral raises would specifically target the side of your shoulders for a more balanced look and strength.

2. Biceps: Rows involve your biceps, but isolated movements like bicep curls are not included.

3. Triceps: Push-ups work your triceps, but exercises like overhead triceps extensions or triceps pushdowns could give them more direct work.

4. Calves: There are no direct calf exercises in the suggested plan. Standing or seated calf raises would target these.

For a beginner, the current plan covers the major compound movements very well. As you get stronger, you might consider adding one or two isolation exercises for these areas.

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# Thank you

